

Test Report

Primap (Web Application)

Document Control

Document Information	
Title	Test Report for Primap (Web Application)
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Version History

Version	Date	Description of Changes
1.0	March 27th, 2026	Initial submission

1. Test Suite Composition

Test Type	Test Count	Proportion (%)	Primary Framework / Tool
Unit Tests	307	92.5%	Vitest + Testing Library
Integration Tests	25	7.5%	Vitest Integration Tests
End-to-End Tests	0	0%	Not Implemented

Rationale

We prioritised unit tests because Primap’s highest-risk business logic, such as role/access control and observation validation/submission rules, is implemented in server actions and utility modules, where tests are fast, deterministic, and isolated.

We then complemented this with a smaller set of integration tests to validate critical cross-module behaviour that unit tests cannot fully guarantee, including routing/middleware flows, data persistence, and side effects such as cancellation notifications.

We did not include E2E tests at this stage, as the team focused on stabilising core business logic and maintaining fast CI feedback cycles. E2E coverage is planned in subsequent iterations as targeted smoke tests for key workflows.

2. Requirements Traceability

Critical Use Case ID	Use Case Name	Validating Test Case ID(s)	Test Type(s)
UC-01	Request Account Creation	TC-UNIT-ACC-REQ-01, TC-INT-AUTH-01	Unit, Integration
UC-03	Sign Up for a Walk Slot	TC-UNIT-WALK-01, TC-UNIT-WALK-02, TC-UNIT-WALK-03	Unit
UC-04	Cancel Walk Slot Participation	TC-UNIT-WALK-04, TC-UNIT-WALK-05, TC-INT-CANCEL-01	Unit, Integration
UC-06	Submit Observation	TC-UNIT-OBS-01, TC-UNIT-OBS-02, TC-UNIT-OBS-03, TC-UNIT-OBS-04	Unit
UC-13	Approve Volunteer Account	TC-UNIT-ADMIN-01, TC-INT-ADMIN-01	Unit, Integration

The mapping shows strong automated coverage for core workflows, particularly walk participation and submission (UC-03, UC-04, UC-06) and the approval process (UC-13). These are supported by integration tests that validate cross-module behaviour, including interactions between server actions and side effects such as notifications.

For UC-01, current tests primarily cover downstream behaviour (e.g., pending state and approval handling) rather than the initial request-submission flow itself, leaving a gap in validating the registration input and creation process directly.

The main limitation is the absence of browser-level E2E validation. As a next step, we plan to introduce a small number of E2E smoke tests targeting critical user journeys, specifically onboarding (UC-01) and observation submission (UC-06), to ensure correctness across the full UI-to-backend flow.

3. Code Coverage and Improvements

Final Coverage Metrics

- Line Coverage: 92.33% (530/574)
- Branch Coverage: 82.78% (303/366)
- Statement Coverage: 92.03% (601/653)
- Function Coverage: 87.93% (102/116)

Action Taken Based on Coverage

The coverage report showed that server actions and utility modules were generally well covered, but it also highlighted several clear gaps.

In particular, `app/api/media/route.ts` and `lib/report-slot-data.ts` currently have no automated coverage. In addition, overall function coverage is 87.93%, which is below the 90% threshold we set.

Based on this, the next testing improvements are to add route-level tests for `app/api/media/route.ts`, covering both successful and error paths, and to add focused unit tests for `lib/report-slot-data.ts`, especially for transformation logic and edge cases. Since browser-level coverage is also currently absent, we also plan to introduce a small number of end-to-end smoke tests for critical volunteer workflows.

4. What Testing Revealed

Bug Identified

From integration tests for our offline sync flow, test case “processOutbox skips items at MAX_RETRIES (5)” failed. The failure output stated that the outbox was empty when it should have still contained the item. We realised we had cleared the outbox completely after processing it. We fixed it by only clearing the outbox of items that have been successfully executed instead. This allowed the test to pass.

Reflection

While writing tests for our saveDraft function, we realised that allowing walk_completion and outcome to potentially be undefined added more test cases to validate and handle these cases properly. We decided that this was unnecessary as we should simply set a default value for these fields instead in each report which makes logical sense as a report should have defined outcomes anyways and reduces the number of edge cases to consider as well.